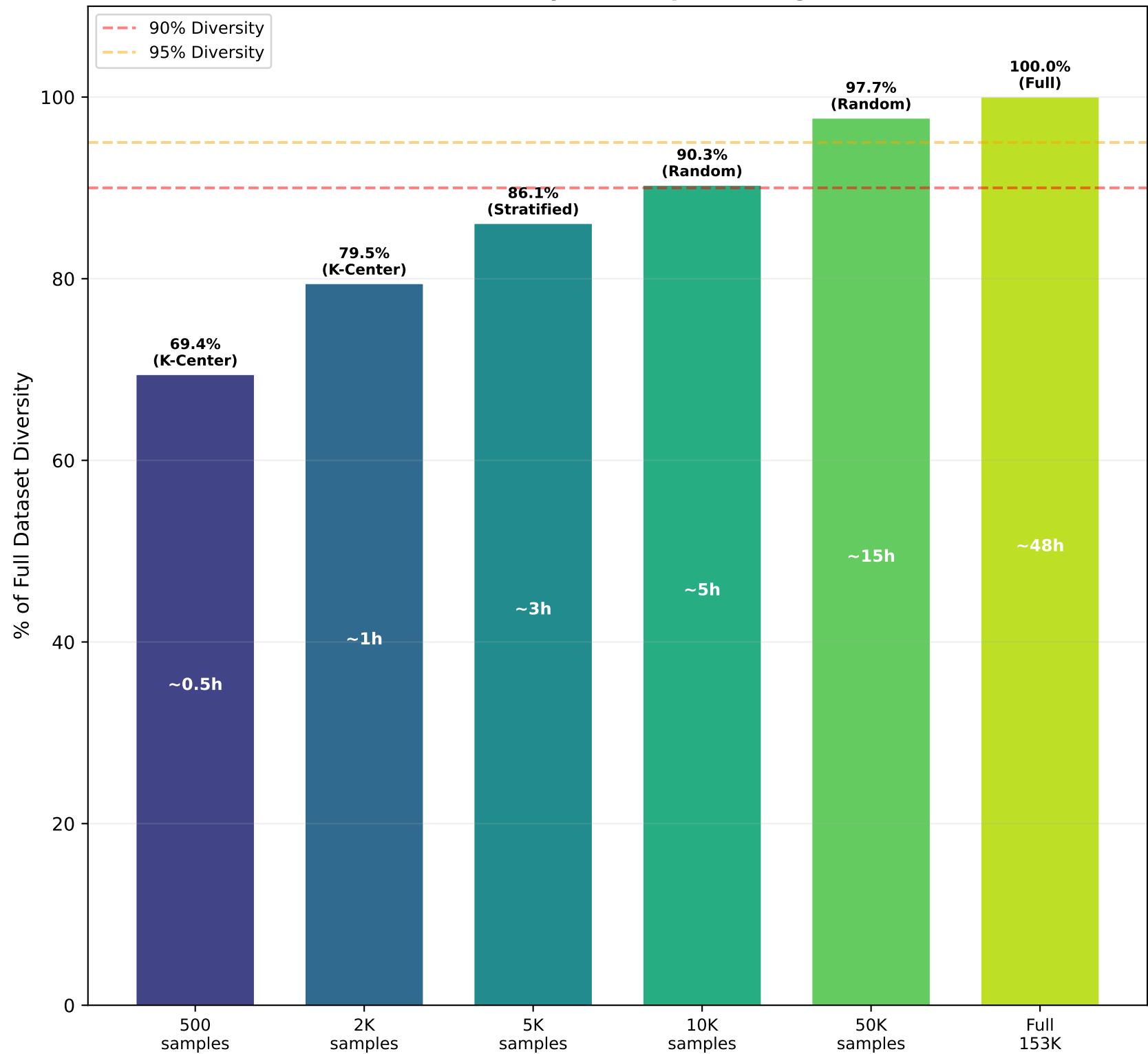


Practical Training Guide: Data Selection for Every Budget

(a) Diversity vs Compute Budget



(b) Recommended Sample Allocation per Budget

Budget	Strategy	Self-Inv	DocQA	Complex	Reasoning	Code	Math	Creative
500	K-Center	100	75	75	75	75	50	50
2K	K-Center	350	300	300	250	300	250	250
5K	Stratified	870	765	685	705	700	650	625
10K	Random (opt)	1740	1530	1370	1410	1400	1300	1250
50K	Random (opt)	8700	7650	6850	7050	7000	6500	6250
Full	All data	4975	21432	54223	17789	15411	11450	9363

Note: "opt" = diversity-optimized ratios (Self-Inv upweighted 3.2%→17.4%).
K-Center-Greedy recommended for N<5K (+33% diversity). Random is near-optimal for N>5K.